

Tutorial Problems on Crystal Structure.

**Crystal Defects, Geometry, and XRD Structure
Determination**

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How to Use This Set

- Attempt each problem before revealing its solution.
- Keep units explicit throughout every calculation.
- For XRD, use $\theta = \frac{1}{2}(2\theta)$.
- Check whether the final result is physically reasonable.

Useful constants

$$k_B = 8.617 \times 10^{-5} \text{ eV K}^{-1}$$

$$1 \text{ \AA} = 10^{-10} \text{ m}$$

$$\lambda_{\text{CuK}\alpha} = 1.5406 \text{ \AA}$$

Example 1: Problem

Equilibrium vacancies

A metal has 8.47×10^{28} atomic sites per cubic metre. The vacancy formation energy is $Q_v = 0.90$ eV/atom.

At $T = 1000$ K, calculate:

1. the equilibrium vacancy fraction, N_v/N ;
2. the number of vacancies per cubic metre;
3. the number of vacancies in 1.00 cm^3 .

Assume independent vacancies and use the dilute-defect approximation.

Example 1: Solution

The equilibrium vacancy relation is

$$\frac{N_v}{N} = \exp\left(-\frac{Q_v}{k_B T}\right).$$

$$\frac{N_v}{N} = \exp\left[-\frac{0.90}{(8.617 \times 10^{-5})(1000)}\right] = 2.91 \times 10^{-5}.$$

Therefore,

$$N_v = (8.47 \times 10^{28})(2.91 \times 10^{-5}) = 2.47 \times 10^{24} \text{ m}^{-3}.$$

Since $1 \text{ cm}^3 = 10^{-6} \text{ m}^3$,

Result

$$N_v(1 \text{ cm}^3) = 2.47 \times 10^{18} \text{ vacancies}$$

Example 2: Problem

ASTM size and Hall-Petch

A steel has ASTM grain-size number $G = 8$. At $100\times$, the ASTM relation is

$$n = 2^{G-1}$$

grains per square inch of micrograph. Approximate every grain by a circle of equal mean area.

Given the Hall-Petch relation

$$\sigma_y = \sigma_0 + k_y d^{-1/2},$$

with $\sigma_0 = 70$ MPa and $k_y = 0.74$ MPa $\sqrt{\text{m}}$, determine the equivalent grain diameter d and the yield strength σ_y .

Example 2: Solution A

At 100 $\times$,

$$n = 2^{8-1} = 128 \text{ grains/in}^2 \text{ of micrograph.}$$

The actual specimen area is reduced by 100², so

$$N_A = 128(100)^2 = 1.28 \times 10^6 \text{ grains/in}^2.$$

The mean area per grain is

$$\bar{A} = \frac{(0.0254)^2}{1.28 \times 10^6} = 5.04 \times 10^{-10} \text{ m}^2.$$

For a circle, $\bar{A} = \pi d^2/4$:

Result

$$d = \sqrt{4\bar{A}/\pi} = 25.3 \times 10^{-6} \text{ m}$$

Example 2: Solution B

Insert $d = 25.3 \times 10^{-6}$ m into Hall-Petch:

$$\begin{aligned}\sigma_y &= 70 + 0.74(25.3 \times 10^{-6})^{-1/2} \\ &= 70 + 147 \\ &= 217 \text{ MPa.}\end{aligned}$$

Result

$$\sigma_y \approx 217 \text{ MPa}$$

Interpretation

ASTM G converts a counted 2D grain population into a characteristic size; Hall-Petch then connects that size to strength. The circular-grain assumption makes d an equivalent diameter.

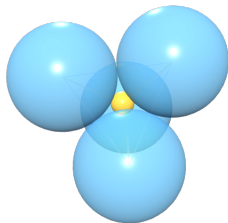
Example 3: Problem

Tetrahedral void

Four identical spheres of radius R touch one another, and their centres form a regular tetrahedron.

Find the radius r of the largest sphere that can occupy the tetrahedral void and touch all four host spheres.

Find r/R and its numerical value.



InteractiveFigure

Example 3: Solution A

The centres of the four host spheres form a regular tetrahedron of edge

$$a_t = 2R.$$

The distance from the centre of a regular tetrahedron to any vertex is

$$R_t = \frac{\sqrt{6}}{4}a_t.$$

The void sphere touches every host sphere, so the same centre-to-vertex distance is

$$R_t = R + r.$$

Hence

$$R + r = \frac{\sqrt{6}}{4}(2R) = \frac{\sqrt{6}}{2}R.$$

Example 3: Solution B

Rearrange the geometric relation:

$$\frac{r}{R} = \frac{\sqrt{6}}{2} - 1.$$

Numerically,

$$\frac{r}{R} = 1.224745 - 1 = 0.224745.$$

Result

$$r = 0.225R$$

Physical meaning

The tetrahedral site is substantially smaller than the host atom. The ratio depends only on geometry, not on the absolute value of R .

Example 4: Problem

Diamond atomic density

Silicon has the diamond cubic structure with lattice parameter

$$a = 0.543 \text{ nm.}$$

Assume hard spheres touch along the nearest-neighbour bond.
Calculate:

1. the number of atoms in the conventional unit cell;
2. the atomic number density in m^{-3} ;
3. the atomic radius r ;
4. the atomic packing factor (APF).

Example 4: Solution A

The conventional diamond cell contains

$$8 \left(\frac{1}{8} \right) + 6 \left(\frac{1}{2} \right) + 4 = 8 \text{ atoms.}$$

Therefore,

$$n = \frac{8}{a^3} = \frac{8}{(0.543 \times 10^{-9})^3} = 4.997 \times 10^{28} \text{ m}^{-3}.$$

The nearest-neighbour distance is $\sqrt{3}a/4 = 2r$, so

$$r = \frac{\sqrt{3}}{8}a = 0.1176 \text{ nm.}$$

Result

$$n \approx 5.00 \times 10^{28} \text{ m}^{-3}, \quad r = 0.1176 \text{ nm}$$

Example 4: Solution B

The atomic packing factor is

$$\begin{aligned}\text{APF} &= \frac{8(4\pi r^3/3)}{a^3} \\ &= \frac{8(4\pi/3)}{a^3} \left(\frac{\sqrt{3}a}{8} \right)^3 \\ &= \frac{\pi\sqrt{3}}{16} = 0.340.\end{aligned}$$

Result

$$\boxed{\text{APF}_{\text{diamond}} = 0.340}$$

Diamond is much more open than FCC, whose APF is 0.740, because tetrahedral bonding fixes a low coordination number of four.

Example 5: Problem

Angle between directions

In a cubic crystal, calculate the angle between the directions

$$[123] \quad \text{and} \quad [1\bar{1}0].$$

Use the direction-vector relation

$$\cos \theta = \frac{u_1 u_2 + v_1 v_2 + w_1 w_2}{\sqrt{u_1^2 + v_1^2 + w_1^2} \sqrt{u_2^2 + v_2^2 + w_2^2}}.$$

For cubic axes, the crystallographic basis vectors are mutually perpendicular and have equal length.

Example 5: Solution

Treat the directions as vectors $\mathbf{a} = (1, 2, 3)$ and $\mathbf{b} = (1, -1, 0)$.

$$\mathbf{a} \cdot \mathbf{b} = 1 - 2 + 0 = -1,$$

$$|\mathbf{a}| = \sqrt{14}, \quad |\mathbf{b}| = \sqrt{2}.$$

Thus

$$\cos \theta = \frac{-1}{\sqrt{28}} = -0.18898, \quad \theta = 100.89^\circ.$$

Result

$$\boxed{\theta \approx 100.9^\circ}$$

If the angle between undirected lines were requested, the acute supplement would be 79.1° .

Example 6: Problem

A common perpendicular

Find a crystallographic direction that is perpendicular to both

$$[111] \quad \text{and} \quad [1\bar{1}0].$$

Express the answer using the smallest integer indices and verify both dot products.

Hint

For two non-parallel vectors, the cross product points perpendicular to both.

Example 6: Solution

Let $\mathbf{a} = (1, 1, 1)$ and $\mathbf{b} = (1, -1, 0)$. Then

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = (1, 1, -2).$$

Check:

$$(1, 1, -2) \cdot (1, 1, 1) = 1 + 1 - 2 = 0,$$

$$(1, 1, -2) \cdot (1, -1, 0) = 1 - 1 + 0 = 0.$$

Result

$[11\bar{2}]$ or the opposite direction $[\bar{1}\bar{1}2]$

Example 7: Problem

Interplanar spacing

A cubic crystal has lattice parameter

$$a = 0.3615 \text{ nm.}$$

Calculate the perpendicular distance between adjacent parallel (211) planes.

For a cubic crystal,

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}.$$

Example 7: Solution

For (211),

$$h^2 + k^2 + l^2 = 2^2 + 1^2 + 1^2 = 6.$$

Therefore,

$$d_{211} = \frac{0.3615}{\sqrt{6}} \text{ nm} = 0.1476 \text{ nm}.$$

Result

$$d_{211} = 0.1476 \text{ nm} = 1.476 \text{ \AA}$$

The spacing is geometric; whether (211) diffracts also depends on the structure factor.

Example 8: Problem

Unknown cubic metal

The first seven powder peaks are measured with Mo $K\alpha$ radiation, $\lambda = 0.7107 \text{ \AA}$:

$$2\theta = 20.20^\circ, 28.72^\circ, 35.36^\circ, 41.06^\circ, \\ 46.17^\circ, 50.87^\circ, 55.28^\circ.$$

Determine the cubic structure, index the peaks, and calculate the lattice parameter.

Assume a single phase and that no low-angle peaks are missing.

Example 8: Solution A

Calculate $q_i = \sin^2 \theta_i$, then divide by q_1 :

$$\frac{q_i}{q_1} \approx 1, 2, 3, 4, 5, 6, 7.$$

The SC sequence omits $N = 7$, while BCC allows

$$N = 2, 4, 6, 8, 10, 12, 14.$$

Thus the peaks index as

$$(110), (200), (211), (220), (310), (222), (321).$$

Every assignment satisfies $h + k + l = 2n$.

Result

BCC

Example 8: Solution B

For a cubic phase,

$$a_i = \frac{\lambda \sqrt{N_i}}{2 \sin \theta_i}.$$

Using the first peak, $N_1 = 2$ and $\theta_1 = 10.10^\circ$:

$$a = \frac{0.7107 \sqrt{2}}{2 \sin 10.10^\circ} = 2.866 \text{ \AA}.$$

Repeating the calculation for all seven peaks gives the same value within rounding.

Result

$$a \approx 2.866 \text{ \AA} = 0.2866 \text{ nm}$$

Example 9: Problem

Identify an FCC pattern

An elemental cubic powder gives the following peaks with Cu $K\alpha$ radiation:

$$2\theta = 43.32^\circ, 50.45^\circ, 74.13^\circ, \\ 89.94^\circ, 95.15^\circ, 116.93^\circ.$$

Determine the structure, index the peaks, and calculate a .

Example 9: Solution A

The normalized values are

$$\frac{\sin^2 \theta_i}{\sin^2 \theta_1} \approx 1, \frac{4}{3}, \frac{8}{3}, \frac{11}{3}, 4, \frac{16}{3}.$$

Multiplying by 3 gives

$$N = 3, 4, 8, 11, 12, 16.$$

These are the FCC families

$$(111), (200), (220), (311), (222), (400).$$

All indices are either all odd or all even.

Result

FCC

Example 9: Solution B

Using the first peak, $N = 3$ and $\theta = 21.66^\circ$:

$$a = \frac{(1.5406 \text{ \AA})\sqrt{3}}{2 \sin 21.66^\circ} = 3.615 \text{ \AA}.$$

The remaining peaks give the same a within the precision of the angles.

Result

$$a \approx 3.615 \text{ \AA} = 0.3615 \text{ nm}$$

Structure check

The observed (200) peak rules out diamond cubic, where (200) is systematically absent.

Example 10: Problem

FCC or diamond?

An elemental cubic material gives Cu $K\alpha$ peaks at

$$2\theta = 28.44^\circ, 47.30^\circ, 56.12^\circ,$$

$$69.13^\circ, 76.38^\circ, 88.03^\circ.$$

Identify the structure, index the peaks, and calculate the lattice parameter. Explain which missing reflection distinguishes it from ordinary FCC.

Example 10: Solution A

Normalize the $\sin^2 \theta$ values:

$$1, \frac{8}{3}, \frac{11}{3}, \frac{16}{3}, \frac{19}{3}, 8.$$

Multiplication by 3 gives

$$N = 3, 8, 11, 16, 19, 24.$$

The indexing is

$$(111), (220), (311), (400), (331), (422).$$

This is the characteristic diamond-cubic sequence.

Result

Diamond cubic

Example 10: Solution B

Using (111),

$$a = \frac{(1.5406 \text{ \AA})\sqrt{3}}{2 \sin 14.22^\circ} = 5.431 \text{ \AA}.$$

For an FCC lattice with this a , (200) would occur near $2\theta = 32.96^\circ$. It is absent because the diamond basis gives

$$1 + i^{h+k+l} = 1 + i^2 = 0$$

for (200).

Result

$$a \approx 5.431 \text{ \AA} = 0.5431 \text{ nm}$$

Key Results

Topic	Essential result
Vacancies	$N_v/N = \exp[-Q_v/(k_B T)]$
Grain size	ASTM $G \rightarrow d \rightarrow \sigma_y = \sigma_0 + k_y d^{-1/2}$
Tetrahedral void	$r/R = \sqrt{6}/2 - 1 = 0.225$
Diamond	8 atoms/cell; APF = $\pi\sqrt{3}/16 = 0.340$
Directions	Dot product: angle; cross product: normal direction
Cubic planes	$d_{hkl} = a/\sqrt{h^2 + k^2 + l^2}$
XRD indexing	$\sin^2 \theta \propto N$, then apply extinction rules

**A valid structure assignment must give both an allowed sequence
and one consistent lattice parameter.**

References

- W. D. Callister Jr. and D. G. Rethwisch, *Materials Science and Engineering: An Introduction*.
- B. D. Cullity and S. R. Stock, *Elements of X-Ray Diffraction*, 3rd ed.
- ASTM E112, *Standard Test Methods for Determining Average Grain Size*.

The ASTM conversion in Example 2 uses an equivalent circular diameter based on mean planar grain area.